gpeyt1-33 · Galvanized coating life

GEO PE · Long Calculation Problems I — Topic 7: Problematic Soil & Rock Conditions

Problem

MSE wall steel strips are hot-dip galvanized with a 86 μm zinc coating. Backfill resistivity tests average 4,200 Ω·cm, pH 7.1, chlorides 60 ppm. Estimate how long the zinc lasts using the AVERAGE galvanized loss constants before the base steel starts to corrode.

Most nearly, the zinc coating life is:

A. 7 years
B. 2 years
C. 5 years
D. 3 years

Given equations / data

> $x = K\,t^{n}$; galvanized: $n = 0.65$, $K = 25$ (avg) / 50 (max) μm

Worked solution

1. 4,200 Ω·cm → moderately corrosive (2,000–5,000)
2. t = (x/K)^(1/n) = (86/25)^(1/0.65) = 3.44^1.538 = 6.7 yr

Final answer

7 years

Why the other values are wrong

- **2 years** — used the maximum K = 50
- **5 years** — divided by n instead of taking the 1/n root
- **3 years** — used carbon-steel constants

References (Jayce's library)

- NCEES PE Civil Reference Handbook v2.1 §3.17.1–3.17.2, pp. 197–198

Original problem inspired by the problem type in: https://www.youtube.com/watch?v=ZOpOWZXWblU (no creator values copied).

Personal use only — draft study material for Jayce Wright. Not for redistribution. Not reviewed by a licensed PE.